Run report: C_4 ladder and ablation, 2026-10-14T20:08Z

Command: `research/gaudry_quartic_d4_cost_20261014/run_ladder.sh` (`RAYON_NUM_THREADS=3`), binary `examples/gaudry_quartic_d4` sha256 `c818a579…f524`, commit `2b49f38d1`, protocol `PROTOCOL.md` (Part A). Host: Xeon @ 2.10 GHz, 4 vCPU, shared with the ecbench session below. Wall times are descriptive and admitted nowhere.

22 configurations ran. Each wrote `raw/<name>.jsonl` (one row per closed solve) and `raw/<name>.log`; none failed. Unsolved systems are counted from the logs ("did not close"): 20 in all, every one listed in `raw/summary.json` per configuration. All 648 closed solves agree with the meet-in-the-middle enumeration (measured). Means and intervals are derived by `examples/gaudry_quartic_d4_summary.rs`: 10,000 bootstrap resamples, seed 1.

configuration	closed / unsolved	mean C_4	95 % interval	first solve
p29 new planted	17 / 3	$5.49 \cdot 10^7$	5.41–5.62	$1.19 \cdot 10^8$
p29 new uniform	297 / 3	$6.23 \cdot 10^7$	6.15–6.34	$1.19 \cdot 10^8$
p29 committed planted	17 / 3	$2.35 \cdot 10^{10}$	2.30–2.39	
p29 committed uniform	50 / 0	$2.36 \cdot 10^{10}$	2.34–2.37	
p269 new planted	19 / 1	$6.90 \cdot 10^7$	6.88–6.92	$1.41 \cdot 10^8$
p269 new uniform	50 / 0	$6.84 \cdot 10^7$	6.83–6.85	$1.40 \cdot 10^8$
p269 committed planted	9 / 1	$2.55 \cdot 10^{10}$	2.54–2.55	
p1009 new planted / uniform	20 / 0, 50 / 0	8.21 / $8.39 \cdot 10^7$		$1.55 \cdot 10^8$
p1009 committed planted	5 / 0	$2.56 \cdot 10^{10}$		
p4049 new planted / uniform	10 / 0, 20 / 0	8.06 / $8.47 \cdot 10^7$		$1.59 \cdot 10^8$
p4049 committed planted	3 / 0	$2.56 \cdot 10^{10}$		

The nine-configuration ablation at $p = 269$ is in the study's `README.md` and `figures/c4_ablation.svg`.

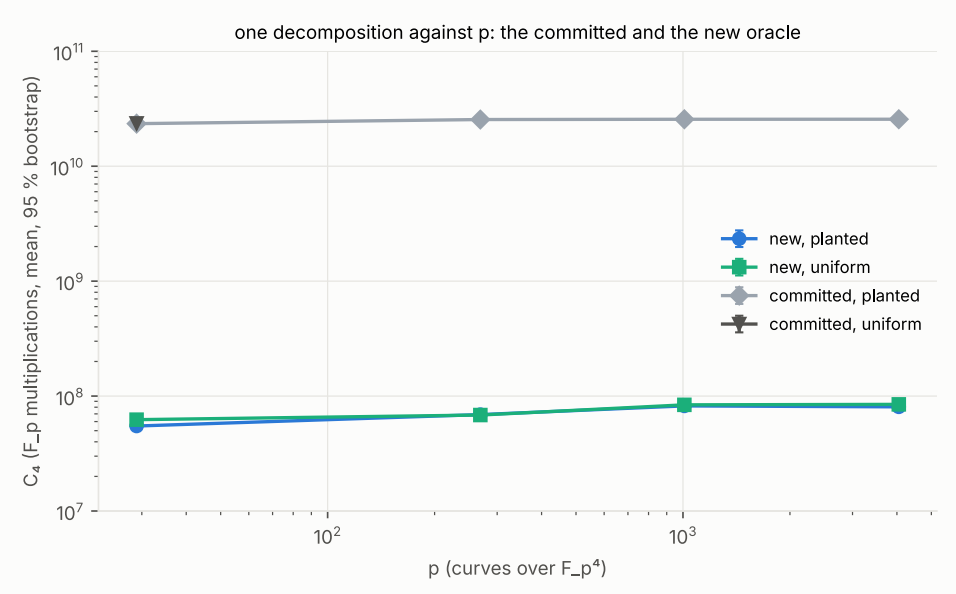


Figure 1: C_4 against p (recorded means and bootstrap intervals).